

```
# Import libraries

import pandas as pd
import numpy as np

# Load dataset

df = pd.read_csv("/content/Social_Network_Ads.csv")

# Check dataset

print(df.head())
print(df.info())

# Check missing values

print(df.isnull().sum())

# Encode Gender

df['Gender'] = df['Gender'].map({'Male':1, 'Female':0})

# Correlation

print(df.corr())

# Feature and target

X = df[['Age', 'EstimatedSalary']]
y = df['Purchased']

# Split data

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0
)

# Train model
```

```
from sklearn.linear_model import LogisticRegression
```

```
logreg = LogisticRegression()
```

```
logreg.fit(X_train, y_train)
```

```
# Predict
```

```
y_pred = logreg.predict(X_test)
```

```
# Accuracy
```

```
from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", accuracy)
```

```
# Error rate
```

```
print("Error Rate:", 1 - accuracy)
```

```
# Confusion matrix
```

```
from sklearn.metrics import confusion_matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("Confusion Matrix:\n", cm)
```

```
# Extract values
```

```
TN, FP, FN, TP = cm.ravel()
```

```
print("TP:", TP)
```

```
print("TN:", TN)
```

```
print("FP:", FP)
```

```
print("FN:", FN)
```

```
# Precision
```

```
print("Precision:", TP / (TP + FP))
```

```
# Recall
```

```
print("Recall:", TP / (TP + FN))
```